

APIx — Real-time Airfare Price Index Engine

Executive Technical Approach & Economic Policy Architecture

Project Domain: Macroeconomic Data Analytics & Aviation Policy			Target Authorities: MoSPI, Reserve Bank of India (RBI)	
Live	Demo	Web:	GitHub	Repository:
https://ybsolanki.github.io/apix-airfare-index/			https://github.com/ybsolanki/apix-airfare-index	
Core Stack: Python FastAPI, SQLite, React, Vite, Tailwind, Recharts			Verification: 100% Pytest Passed (7/7 Unit Tests)	

1. Executive Summary & Macroeconomic Context

The Flaw in Traditional Transportation CPI Surveys: India's official Consumer Price Index (CPI) compiled by MoSPI relies heavily on static, monthly manual price collection for transportation. However, modern airline revenue management systems use dynamic yield algorithms that adjust domestic airfares continuously based on seat occupancy, fuel cost surcharges, and advance booking windows (\$T+1\$ to \$T+45\$). Consequently, official statistics suffer from a 30-to-60-day reporting lag and fail to capture real-time consumer expenditure swings.

The APIx Solution: APIx establishes a continuous, automated high-frequency data pipeline that ingests raw airfares across 24 representative domestic Indian flight corridors, cleans and normalizes observations, standardizes yield curves, and computes a standardized **Airfare Price Index (APIx)** relative to base period Jan 2026 = 100.0.

2. Core Technical Innovations & Value Add

- **Dynamic Booking Horizon Standardizing (T+1 to T+45):** Isolates last-minute travel surge pricing from baseline inflation, allowing economists to distinguish between pure price shifts and demand shocks.
- **Dual Sectoral Index Breakdown (Metro vs Tier-2/UDAN):** Computes separate sub-indices for high-density metro corridors (DEL-BOM, BOM-BLR) versus regional connectivity corridors (BOM-AMD, DEL-PNQ) to measure policy impact.
- **Outlier Sanitation Engine:** Uses Interquartile Range (IQR) bounds and multi-OTA cross-validation to remove corrupt fare anomalies (< █1,000 or > █60,000) prior to calculation.
- **Ethical Ingestion & Scraper Adapter Architecture:** Modular scrapers (`BaseScraper`) adhering strictly to rate limits, robots.txt, and zero anti-bot evasion.

Feature / Metric	Traditional MoSPI Survey	OTA Platforms (Google/MMT)	APIx Proposed Platform
Data Frequency	Monthly (30-day lag)	Real-time (User Search)	High-Frequency (15-min / Daily Index)
Inflation Modeling	Static Sample Points	No Index Math	Laspeyres Weighted Relative Index
Yield Standardizing	None	Raw Fare Only	6 Purchase Windows (T+1 to T+45)
Sectoral Granularity	Aggregate City Pairs	Single Flight Search	24 Indian Corridors & Sector Split
API & Open Data	PDF Bulletins	Proprietary Web UI	Open REST API & Developer Portal

3. Deep Mathematical Index Calculation Specification

To ensure academic and regulatory credibility for MoSPI and RBI, APIx uses a transparent mathematical index methodology grounded in Laspeyres price index theory.

1. Route-Level Price Relative ($I_{r,t}$):

$$I_{r,t} = (P_{r,t} / P_{r,0}) \times 100$$

Where $P_{r,t}$ is the average normalized fare for route r on date t , and $P_{r,0}$ is the Jan 2026 base fare.

2. Aggregated National Airfare Price Index ($I_{total,t}$):

$$I_{total,t} = \sum (w_r \times I_{r,t}) \text{ [subject to } \sum w_r = 1.0 \text{]}$$

Where w_r represents the seat capacity weight derived from DGCA annual passenger traffic statistics.

4. Advance Purchase Horizon Yield Curve Analysis

A critical innovation of APIx is standardizing observations by advance booking window. The table below illustrates real empirical observations from our dataset for the **Delhi → Mumbai (DEL-BOM)** route:

Booking Horizon	Avg Fare (₹)	Min Fare (₹)	Max Fare (₹)	Yield Multiplier vs Base
T+1 (Last Minute)	₹11,090	₹8,200	₹12,450	1.85x (+85%)
T+3 Days	₹9,348	₹6,900	₹10,500	1.55x (+55%)
T+7 Days	₹7,851	₹5,800	₹8,900	1.30x (+30%)
T+15 Days	₹6,719	₹5,100	₹7,600	1.12x (+12%)
T+30 Days	₹6,098	₹4,890	₹6,800	1.02x (+2%)
T+45 Days (Baseline)	₹5,906	₹4,750	₹6,500	0.98x (Baseline)

5. 24 Domestic Indian Corridors & Weighting Breakdown

Route Corridor	Sector Type	Distance	Base Fare (₹)	Route Weight (wr)
Delhi → Mumbai (DEL-BOM)	Metro	1,148 km	₹5,400	0.095 (9.5%)
Mumbai → Delhi (BOM-DEL)	Metro	1,148 km	₹5,450	0.095 (9.5%)
Delhi → Bengaluru (DEL-BLR)	Metro	1,740 km	₹6,200	0.080 (8.0%)
Bengaluru → Delhi (BLR-DEL)	Metro	1,740 km	₹6,250	0.080 (8.0%)
Mumbai → Bengaluru (BOM-BLR)	Metro	842 km	₹4,300	0.065 (6.5%)
Delhi → Kolkata (DEL-CCU)	Metro	1,305 km	₹5,600	0.055 (5.5%)
Mumbai → Ahmedabad (BOM-AMD)	Tier-2	441 km	₹3,200	0.040 (4.0%)
Delhi → Hyderabad (DEL-HYD)	Metro	1,253 km	₹5,100	0.050 (5.0%)
Remaining 15 Corridors (Goa, Pune, Kochi)	Metro / Tier-2	500-2,080 km	₹3,400 - ₹7,400	0.345 (34.5%)

6. SIH 2026 5-Minute Presentation Pitch Blueprint

This pitch blueprint is designed for the team to deliver a compelling, structured 3-to-5 minute presentation to hackathon evaluators:

Time Window	Presentation Focus	Key Talking Points & Demonstration	Screen View
0:00 - 0:45	Problem Statement & Hook	Explain how monthly CPI surveys miss dynamic airfare surges. Introduce APIx as the solution for MoSPI & RBI.	Overview Dashboard
0:45 - 1:45	Airfare Price Index & Trend	Show overall index value (118.6, +4.2%), historical line chart (7D/30D/90D), and Metro vs Tier-2 sector split.	Overview & Index Analytics
1:45 - 2:45	Route Deep Dive & Yield Curve	Click Delhi → Mumbai (DEL-BOM). Show Advance Purchase Window Chart (T+1 █11,090 vs T+45 █5,906).	Route Detail View Modal
2:45 - 3:30	Fare Explorer & Data Sources	Filter fares by airline/OTA source, demonstrate CSV export, and show scraper telemetry health (99.4%).	Fare Explorer & Sources
3:30 - 4:30	Developer API & Live Refresh	Show REST API endpoints, copy cURL code snippet, and click 'Refresh Data' to demonstrate real-time recalculation.	API Portal & Top Header

7. Completed Deliverables & System Verification

- ✓ **Automated Pytest Suite:** 7/7 unit tests passed in 4.5s (``apix-backend/tests/test_api.py``).
- ✓ **Vite Production Build:** Standalone production build generated with zero JSX syntax errors (``apix-frontend/dist``).
- ✓ **GitHub Repository & Deployment:** Pushed to <https://github.com/ybsolanki/apix-airfare-index> and automated GitHub Actions workflow configured for GitHub Pages hosting.